

Do More AI Agents Review Code Better? *Only If You Can Verify Them.*

A pre-registered comparison of multi-agent LLM code review — and the verifier ablation that changes the answer.

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Motivation

- AI code reviewers increasingly chain **multiple LLM agents** — managers, specialists, voters — and price accordingly.
- We built a controlled ladder of **4 architectures** (single-pass → 3 parallel → manager+specialists → debate&vote): one frozen model, 100 real PRs, pre-registered on OSF.
- Pilot verdict: teams lose.** They find more real issues but drown them in false alarms (F1 0.37–0.41 vs 0.49 single-pass).

So is multi-agent review hopeless — or just missing a gatekeeper?

A Ladder of Verifiers

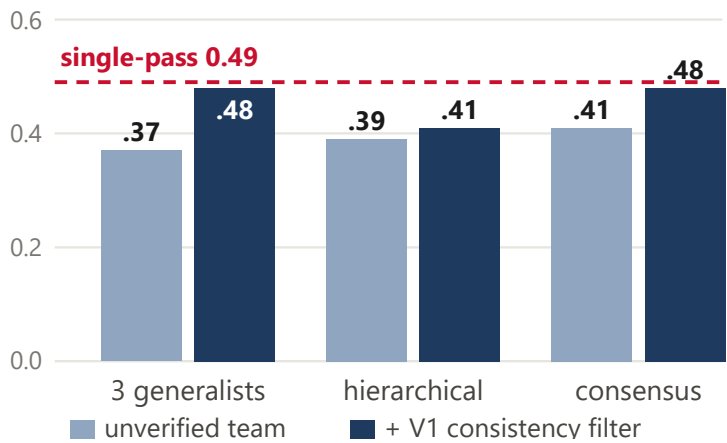
Keep the agent teams; add **verifiers of increasing strength** over their pooled findings:

V0	Raw union everything every agent said	noise
V1	Self-consistency keep only findings ≥2 independent runs agree on — free	works
V2	LLM judge (yes/no) a strong model asked "is this real?"	rubber stamp
V3	Structured judge rubric × 3 criteria × repeated scoring	signal, no F1

Held constant: frozen model & prompts, same 21-PR batch × 3 runs, one evaluator. **Non-circular:** generator, match-judge, and verifier are three different model families.

Key Results

Figure 1. A free consistency check rescues the agent teams (semantic F1).



- The free verifier rescues the teams** (+0.11 F1) to parity with single-pass; consensus keeps **+6pt recall at equal F1** — the win when misses cost more.
- Smart judges aren't verifiers:** a strong LLM asked yes/no approved 82% of everything; rubric-structured scoring discriminates (AUC 0.68) but still can't beat the free check.

Surprise: the answer key was only ~55% complete. Agent teams surfaced **88 real issues** the human reviews never listed — much of their "imprecision" was the benchmark's fault.

Why It Matters

- Never deploy multi-agent review without a consistency filter** — unverified, extra agents buy noise at 3–5× the cost.
- Verification structure beats judge intelligence:** independent agreement outperformed a frontier-model judge.
- Benchmark golden sets under-credit high-recall tools** — evaluate completeness before trusting precision.

Next Steps

- Run the **pre-registered confirmatory campaign** (100 PRs × 4 arms × 3 runs; gated on compute quota).
- Tool-grounded verifiers** — execution & static analysis as code's "proof checker".
- Open science:** all runs & ablations released for replay.

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